Area, Surface Area & Volume

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Sine, cosine, and tangent ratios**

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

Final Answer: $x = 1$ and $x = 3$. Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

2 **[Foundational]** • **Target: Sine, cosine, and tangent ratios**

A data set recorded for sine, cosine, and tangent ratios consists of the four values $\{18, 22, 26, 32\}$.

What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 24.5. The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

3 **[Foundational]** • **Target: Sine, cosine, and tangent ratios**

For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(3)$.

Final Answer: 23. Substitute $x = 3$: $f(3) = 6(3) + 5 = 23$.

4 **[Foundational]** • **Target: Sine, cosine, and tangent ratios**

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

Final Answer: $x = 2$ and $x = 4$. Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

5 **[Foundational]** • **Target: Sine, cosine, and tangent ratios**

For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(4)$.

Final Answer: 29. Substitute $x = 4$: $f(4) = 6(4) + 5 = 29$.

6 **[Foundational]** • **Target: Sine, cosine, and tangent ratios**

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

Final Answer: $x = 3$ and $x = 5$. Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

7 **[Foundational]** • **Target: Sine, cosine, and tangent ratios**

For the model $f(x) = 6x + 5$ in sine, cosine, and tangent ratios, evaluate $f(5)$.

Final Answer: 35. Substitute $x = 5$: $f(5) = 6(5) + 5 = 35$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Sine, cosine, and tangent ratios

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

Final Answer: $x = 4$ and $x = 6$. Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

9 [Intermediate] • Target: Solving for sides in right triangles

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 85°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (50^\circ + 45^\circ) = 85^\circ$.

10 [Intermediate] • Target: Solving for sides in right triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 8 cm. Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

11 [Intermediate] • Target: Solving for sides in right triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 9. Substituting $x = 1$ into $y = 3x + 6$ gives $y = 3(1) + 6 = 9$.

12 [Intermediate] • Target: Solving for sides in right triangles

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 78°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (50^\circ + 52^\circ) = 78^\circ$.

13 [Intermediate] • Target: Solving for sides in right triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 12 cm. Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

14 [Intermediate] • Target: Solving for sides in right triangles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 12. Substituting $x = 2$ into $y = 3x + 6$ gives $y = 3(2) + 6 = 12$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 2.01 Lines & Angles

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 71°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (50^\circ + 59^\circ) = 71^\circ$.

16 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

17 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 15. Substituting $x = 3$ into $y = 3x + 6$ gives $y = 3(3) + 6 = 15$.

18 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In ΔABC studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 64°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (50^\circ + 66^\circ) = 64^\circ$.

19 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 20 cm. Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

20 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 7.01 Circles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 18. Substituting $x = 4$ into $y = 3x + 6$ gives $y = 3(4) + 6 = 18$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 45^\circ) = 88^\circ$.

22 [Foundational] • Target: The Pythagorean theorem and its converse

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

23 [Foundational] • Target: The Pythagorean theorem and its converse

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 11

b. 17

c. 15 ✓

d. 19

Substituting $x = 3$ into $y = 4x + 3$ gives $y = 4(3) + 3 = 15$.

24 [Foundational] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 67° ✓

b. 77°

c. 59°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 66^\circ) = 67^\circ$.

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25 [Foundational] • Target: Extension: Special right triangles

In ΔABC studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 87° ✓

b. 97°

c. 79°

d. 132°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (48^\circ + 45^\circ) = 87^\circ$.

26 [Foundational] • Target: Extension: Special right triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

27 [Foundational] • Target: Extension: Special right triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 14

b. 21

c. 19 ✓

d. 24

Substituting $x = 3$ into $y = 5x + 4$ gives $y = 5(3) + 4 = 19$.

28 [Foundational] • Target: Extension: Special right triangles

In ΔABC studying extension: special right triangles, $m\angle A = 48^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 66° ✓

b. 76°

c. 58°

d. 132°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (48^\circ + 66^\circ) = 66^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: The Pythagorean theorem and its converse

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

30 [Intermediate] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 52^\circ) = 81^\circ$.

31 [Intermediate] • Target: The Pythagorean theorem and its converse

In ΔABC studying the pythagorean theorem and its converse, $m\angle A = 47^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 74° ✓

b. 84°

c. 66°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 59^\circ) = 74^\circ$.

32 [Intermediate] • Target: The Pythagorean theorem and its converse

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

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33 [Intermediate] • Target: Extension: Special right triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

34 [Intermediate] • Target: Extension: Special right triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 9

b. 16

c. 14 ✓

d. 19

Substituting $x = 2$ into $y = 5x + 4$ gives $y = 5(2) + 4 = 14$.

35 [Intermediate] • Target: Extension: Special right triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

36 [Intermediate] • Target: Extension: Special right triangles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 19

b. 26

c. 24 ✓

d. 29

Substituting $x = 4$ into $y = 5x + 4$ gives $y = 5(4) + 4 = 24$.

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37 [Intermediate] • Target: Solving for sides in right triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$?

a. 9

b. 14

c. 12 ✓

d. 15

Substituting $x = 2$ into $y = 3x + 6$ gives $y = 3(2) + 6 = 12$.

39 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

40 [Challenging] • Target: Solving for sides in right triangles • Cross-Lesson: 7.01 Circles

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 15

b. 20

c. 18 ✓

d. 21

Substituting $x = 4$ into $y = 3x + 6$ gives $y = 3(4) + 6 = 18$.

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41 [Challenging] • Target: Solving for angles in right triangles • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 13

c. 11 ✓

d. 15

Substituting $x = 1$ into $y = 4x + 7$ gives $y = 4(1) + 7 = 11$.

42 [Challenging] • Target: Solving for angles in right triangles • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC studying solving for angles in right triangles, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70° ✓

b. 80°

c. 62°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 59^\circ) = 70^\circ$.

43 [Challenging] • Target: Solving for angles in right triangles • Cross-Lesson: 10.01 Coordinate Geometry

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

44 [Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 11.01 Probability

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 15

c. 13 ✓

d. 18

Substituting $x = 1$ into $y = 5x + 8$ gives $y = 5(1) + 8 = 13$.

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45 [Challenging] • Target: Angles of elevation and depression • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 13

b. 20

c. 18 ✓

d. 23

Substituting $x = 2$ into $y = 5x + 8$ gives $y = 5(2) + 8 = 18$.

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